INCH-POUND
MIL-PRF-83530/1C
w/Amendment 3
27 June 2022
SUPERSEDING
MIL-PRF-83530/1C
w/Amendment 2
25 July 2017

PERFORMANCE SPECIFICATION

RESISTOR, VOLTAGE SENSITIVE (VARISTOR)
STYLE RVS10

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall
consist of this specification sheet and MIL-PRF-83530.

Inactive for new design
after 31 March 2008

Parts manufactured prior to the above date
may only be used for replacement purposes.
See QPDSIS-83530 071212
Parts manufactured after above date
shall not be supplied to this Specification.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RVS10 varistors.

1.2 Part or Identifying Number (PIN). Varistors covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded number. The PIN consists of the following:

M83530/1	-	2200D
Detail specification number		Coded number

The coded number will be derived in accordance with MIL-PRF-83530.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, Ohio 43218-3990 or by email resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-83530	-	Resistor, Voltage Sensitive (Varistor Metal -Oxide), General Specification for.

(Copies of these documents are available online at https://quicksearch.dla.mil/).

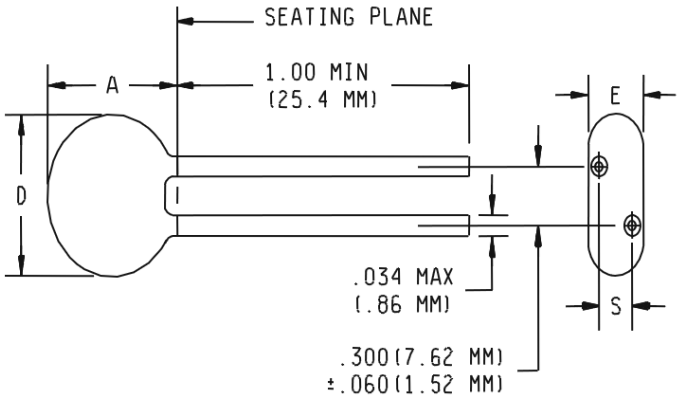
2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-83530.

3.2 Interface and physical dimensions. Varistors shall meet the interface and physical dimensions specified on figure 1, as applicable.

3.3 Nominal varistor voltage. The nominal varistor voltages applicable to this specification shall be as specified in table I. For varistors operated at ambient temperatures in excess of 85°C, the nominal varistor voltage shall be derated in accordance with figure 2.



A		D		E		S			
Max		Max		Max		Min		Max	
Inch	mm	Inch	mm	Inch	mm	Inch	mm	Inch	mm
1.10	27.94	0.95	24.13	0.32	8.13	0.054	1.37	0.26	6.60

FIGURE 1. Style RVS10 varistors, dimensions and configuration.

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3.4 Voltage tolerance. The nominal varistor voltage tolerances applicable to this specification shall be as specified in table I.

3.5 Voltage rating. The voltage rating applicable to parts covered by this specification shall be as specified in table I. For varistors operated at ambient temperatures in excess of 85°C, the voltage shall be derated in accordance with figure 2.

3.6 Energy rating. The energy rating applicable to parts covered by this specification shall be as specified in table I. For varistors operated at ambient temperatures in excess of 85°C, the voltage shall be derated in accordance with figure 2.

TABLE I. Rating and characteristics.

PIN 1/	Nominal varistor voltage (V)	Tolerance (%)		Voltage rating (V) rms	Voltage rating (V) dc	Energy rating (joules)	Clamping voltage at 100 A (V)	Capacitance at 1 MHz (pF)	Clamping voltage at peak current rating (V)
M83530/1-2000B	200	±10		130	175	50	325	3800	570
M83530/1-2200D	220	+10	-5	150	200	55	360	3200	650
M83530/1-4300E	430	+5,	-10	275	369	100	680	1800	1200
M83530/1-5100E	510			320	420	120	810	1500	1450

1/ Parts manufactured prior to 30 March 2008 may only be used for replacement purposes. Parts manufactured after 30 March 2008 shall not be supplied to this Specification.

3.7 Clamping voltage. The varistors shall not exceed the maximum clamping voltage value specified in table I. For varistors operated at ambient temperatures in excess of 85°C, the voltage shall be derated in accordance with figure 2.

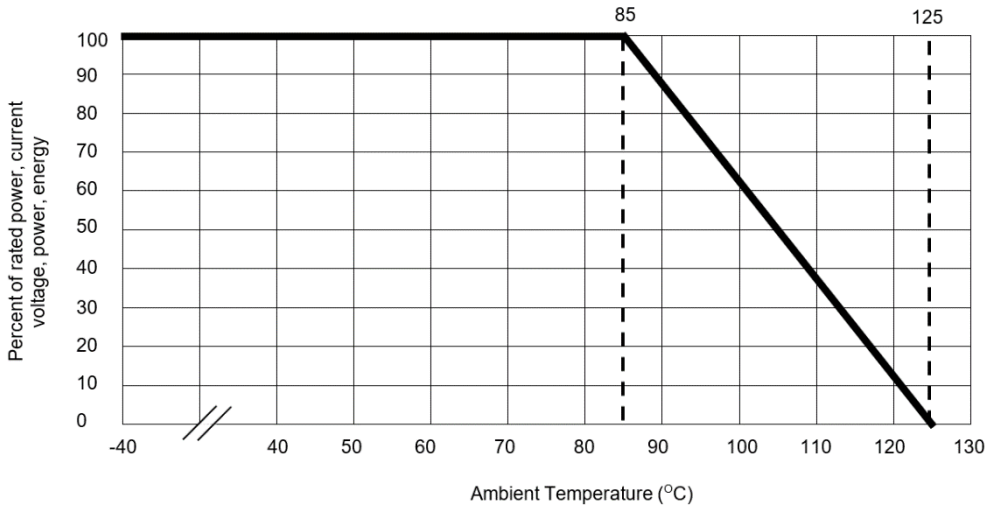


FIGURE 2. Current voltage, power, energy rating versus temperature.

3.8 Capacitance. The varistors shall not exceed the maximum capacitance value specified in table I.

3.9 Peak current rating. The peak current rating applicable to parts covered by this specification shall be 6,000 amperes (A). The clamping voltage at this rating shall not exceed the maximum value specified in table I. For varistors operated at ambient temperatures in excess of 85°C, the voltage shall be derated in accordance with figure 2.

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3.10 Pulse life rating. The pulse life rating applicable to parts covered by this specification shall be 10,000 impulses.

3.11 Power rating. The average power dissipation rating applicable to parts covered by this specification shall be 1.0 W at 85°C. For varistors operated at ambient temperatures in excess of 85°C, the voltage shall be derated in accordance with figure 2.

3.12 Weight. Varistors applicable to this specification shall not exceed a weight of 0.35 ounce (10 grams).

3.13 Pure tin. The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of resistor components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.3).

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-83530.

4.2 Energy. The test peak current shall be 140 A.

4.3 Clamping voltage. The test current shall be 100 A.

4.4 Pulse life. The test current shall be 250 A.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military services system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-83530 are applicable to this specification.

6.2 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.3 Amendment notification. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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